

Companion Results Appendix: GGUF Throughput Prediction

Protocol-complete two-host evidence for the ICASSP manuscript

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1 Scope and cohort

The source manifest contains 23 GGUF files. Across two hosts, the scored data contain 21 unique files and 33 host–file configurations (27 training and 6 held out). Each configuration has one decode and one prefill observation at each of the 3 measured context depths, for 99 decode and 99 prefill rows. The final protocol selector retains 216 observations from 22 unique files in total: 198 scored rows and 18 reference-probe rows excluded from prediction fits and scores. Probe exclusion is host-specific, so a file used as a probe on one host can be scored on the other. All exported rows carry the declared protocol fields. The retry decision, however, uses the worst within-run *decode* CV in an invocation. Prefill rows inherit that invocation’s protocol metadata but are not gated on their own CV. This distinction matters on RTX 5080, where the noisiest prefill row has 39.19% CV even though the associated decode-gate CV is 1.35%.

Host	Backend	Train cfg.	Test cfg.	Scored decode	Scored prefill
MacBook M4 Max	BLAS,MTL	15	4	57	57
RTX 5080	CUDA	12	2	42	42

Host	Retried cells	Gate CV med.	Gate CV max	PP CV med.	PP CV max	PP > 3%
MacBook M4 Max	3	1.43	3.04	1.41	4.84	7/57
RTX 5080	3	0.88	1.65	3.56	39.19	29/42

Attempted configurations with no successful row are MacBook M4 Max: `gpt-oss-120b-MXFP4.gguf`. On the MacBook Pro M4 Max, two raw attempts of the one `gpt-oss-120B` configuration report failed to decode prompt batch, `res=-3`; these are prompt-batch failures, not demonstrated model-load or out-of-memory failures. Host-specific calibration probes are MacBook M4 Max: `Qwen3.8-27B-Q8_0.gguf`, `Qwen3.5-9B-Q8_0.gguf`, `Qwen3.5-4B-Q8_0.gguf`; RTX 5080: none.

1.1 Cohort accounting

Raw records, superseded legacy rows, repeated protocol attempts, selected observations, and scoring exclusions are separated below. A protocol duplicate is a successful protocol row removed by the atomic host–model–phase–depth selector; it is not an additional scored observation.

Host	Raw	Failed	Legacy ok	Protocol ok	Dup. removed	Selected	Probes	Scored
MacBook M4 Max	278	2	132	144	12	132	18	114
RTX 5080	84	0	0	84	0	84	0	84

1.2 Excluded reference-probe observations

These rows complete the 216-observation selected cohort and make the descriptive matched-host and quantization comparisons reproducible. They are reported only as measurements, never as prediction errors.

Table 1: Selected protocol-complete reference-probe observations excluded from prediction fitting and scoring. Measured throughput and SD are in tokens/s; IDs resolve through Table 2.

Host	ID	Phase	Depth	Measured	SD	CV %	Attempts
MacBook M4 Max	M09	decode	0	77.275	0.201	0.260	1
MacBook M4 Max	M09	decode	4,096	74.687	0.153	0.204	1

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Table 1: Selected protocol-complete reference-probe observations excluded from prediction fitting and scoring. Measured throughput and SD are in tokens/s; IDs resolve through Table 2. (continued)

Host	ID	Phase	Depth	Measured	SD	CV %	Attempts
MacBook M4 Max	M09	decode	16,384	67.270	0.341	0.507	1
MacBook M4 Max	M09	prefill	0	1553.086	1.682	0.108	1
MacBook M4 Max	M09	prefill	4,096	1416.098	1.799	0.127	1
MacBook M4 Max	M09	prefill	16,384	992.403	10.398	1.048	1
MacBook M4 Max	M11	decode	0	46.499	0.380	0.818	2
MacBook M4 Max	M11	decode	4,096	45.340	0.169	0.372	2
MacBook M4 Max	M11	decode	16,384	42.119	0.938	2.226	2
MacBook M4 Max	M11	prefill	0	865.397	0.661	0.076	2
MacBook M4 Max	M11	prefill	4,096	784.470	4.299	0.548	2
MacBook M4 Max	M11	prefill	16,384	560.769	14.295	2.549	2
MacBook M4 Max	M13	decode	0	14.902	0.054	0.365	1
MacBook M4 Max	M13	decode	4,096	14.433	0.017	0.118	1
MacBook M4 Max	M13	decode	16,384	13.575	0.092	0.680	1
MacBook M4 Max	M13	prefill	0	238.511	7.257	3.043	1
MacBook M4 Max	M13	prefill	4,096	201.115	2.027	1.008	1
MacBook M4 Max	M13	prefill	16,384	161.599	1.227	0.759	1

Probe-exclusion sensitivity. The three Mac Q8 files defined the historical LLM-reference diagnostic. Their exclusion was fixed before the final device-copy analysis and is retained to avoid a post-hoc cohort change. If they are included and B2 is refit, MAPE is 10.46% over 54 training rows and 13.11% over 12 held-out rows; the latter is unchanged because all probes are Q8 training files while the test files are Q4. On the original 45 non-probe training rows under this refit, MAPE is 11.06%.

2 Aggregate prediction results

Results are separated by host because fitted efficiency factors are host-specific; a pooled row would weight the unequal host cohorts and is not a cross-system score.

Host	Predictor	Split	n	MAPE (%)	Median (%)	P90 / max (%)
MacBook M4 Max	B0 unfitted (eta=1, total)	train	45	33.57	19.37	76.47 / 123.27
MacBook M4 Max	B0 unfitted (eta=1, total)	test	12	46.89	49.97	83.45 / 84.22
MacBook M4 Max	B1 fitted, total params	train	45	15.80	3.20	70.29 / 76.77
MacBook M4 Max	B1 fitted, total params	test	12	49.36	54.66	81.90 / 82.74
MacBook M4 Max	B2 fitted, active params (ours)	train	45	10.39	2.12	26.32 / 78.49
MacBook M4 Max	B2 fitted, active params (ours)	test	12	13.11	8.98	36.65 / 49.92
RTX 5080	B0 unfitted (eta=1, total)	train	36	20.27	14.47	44.15 / 71.19
RTX 5080	B0 unfitted (eta=1, total)	test	6	41.65	40.39	81.22 / 81.95
RTX 5080	B1 fitted, total params	train	36	9.85	8.88	20.95 / 31.73
RTX 5080	B1 fitted, total params	test	6	51.85	50.81	84.50 / 85.11
RTX 5080	B2 fitted, active params (ours)	train	36	10.12	9.34	20.95 / 31.73
RTX 5080	B2 fitted, active params (ours)	test	6	36.15	25.79	62.41 / 68.96
MacBook M4 Max	P1 eta_p per host	train	15	6.81	5.36	16.58 / 33.40
MacBook M4 Max	P1 eta_p per host	test	4	22.68	19.16	44.95 / 51.25
MacBook M4 Max	P2 eta_p per host x quant	train	15	4.13	0.00	15.06 / 25.99
MacBook M4 Max	P2 eta_p per host x quant	test	4	18.68	14.90	36.90 / 42.85
RTX 5080	P1 eta_p per host	train	12	14.46	9.15	31.38 / 52.16
RTX 5080	P1 eta_p per host	test	2	124.51	124.51	207.24 / 227.93
RTX 5080	P2 eta_p per host x quant	train	12	5.86	0.00	21.31 / 25.36
RTX 5080	P2 eta_p per host x quant	test	2	108.18	108.18	184.89 / 204.07

Decode summaries use all three context depths. Prefill headline rows use only depth 0, because the stated prefill model has no existing-prefix term; deeper rows are scope diagnostics below.

2.1 Leave-one-host-out transfer

B2 transfers the source host’s median per-quantization coefficients and substitutes the untouched target host’s bandwidth. The target set contains all eligible rows on that host; because most files also occur on the source host, this isolates host/runtime transfer rather than a simultaneous model-and-hardware holdout. Q4_K_M is shared; the Mac Q4_K test file uses the source-wide median fallback. The second variant is the rejected absolute-time output-projection extension.

Variant	Target host	Source n	Target n	MAPE	Median	Max
B2: all target rows	MacBook M4 Max	36	57	20.82	15.41	75.81
B2: all target rows	RTX 5080	45	42	21.69	18.75	71.53
B2: target test rows	MacBook M4 Max	36	12	13.94	8.63	47.67
B2: target test rows	RTX 5080	45	6	36.70	26.18	71.53
rejected two-term	MacBook M4 Max	36	57	57.06	54.22	78.57
rejected two-term	RTX 5080	45	42	116.85	98.66	349.25

For context, target-fitted B2 gives all-row MAPE of 10.96% on Mac and 13.84% on RTX. Unlike the untouched-target transfer rows, these comparators mix training rows used to fit the target coefficients with held-out rows.

Host	Split	Architecture	n	B0	B1	B2
MacBook M4 Max	train	dense	36	23.21	7.14	5.66
MacBook M4 Max	train	MoE	9	74.98	50.45	29.29
MacBook M4 Max	test	dense	6	11.18	17.75	3.28
MacBook M4 Max	test	MoE	6	82.61	80.97	22.95
RTX 5080	train	dense	33	15.73	10.57	10.57
RTX 5080	train	MoE	3	70.28	1.95	5.18
RTX 5080	test	dense	3	3.54	20.40	20.40
RTX 5080	test	MoE	3	79.77	83.31	51.90

2.2 Per-layer KV correction

With each variant refit on training rows, replacing a uniform-global KV denominator by the frozen per-layer metadata produces:

Host	Split	n	Uniform KV MAPE (%)	Per-layer KV MAPE (%)
MacBook M4 Max	train	45	12.68	10.39
MacBook M4 Max	test	12	21.23	13.11
RTX 5080	train	36	12.37	10.12
RTX 5080	test	6	37.87	36.15

2.3 Decode context behavior

All 33 scored host–file configurations slow monotonically from 0 through 16,384 tokens. The normalized rate is each model’s throughput divided by its own depth-0 throughput.

Host	Depth	Train n	Test n	Train B2 MAPE	Test B2 MAPE	Median normalized rate
MacBook M4 Max	0	15	4	11.30	17.86	1.000
MacBook M4 Max	4,096	15	4	7.59	12.83	0.945
MacBook M4 Max	16,384	15	4	12.27	8.63	0.837
RTX 5080	0	12	2	10.19	44.52	1.000
RTX 5080	4,096	12	2	4.49	38.28	0.985
RTX 5080	16,384	12	2	15.67	25.64	0.922

2.4 Prefill scope diagnostic

Depth 0 is the primary prefill fit and score. The same fitted coefficients are applied unchanged at larger existing-prefix depths. These errors are prediction errors, distinct from within-run prefill variability; the RTX held-out result is poor even before that distinction is considered.

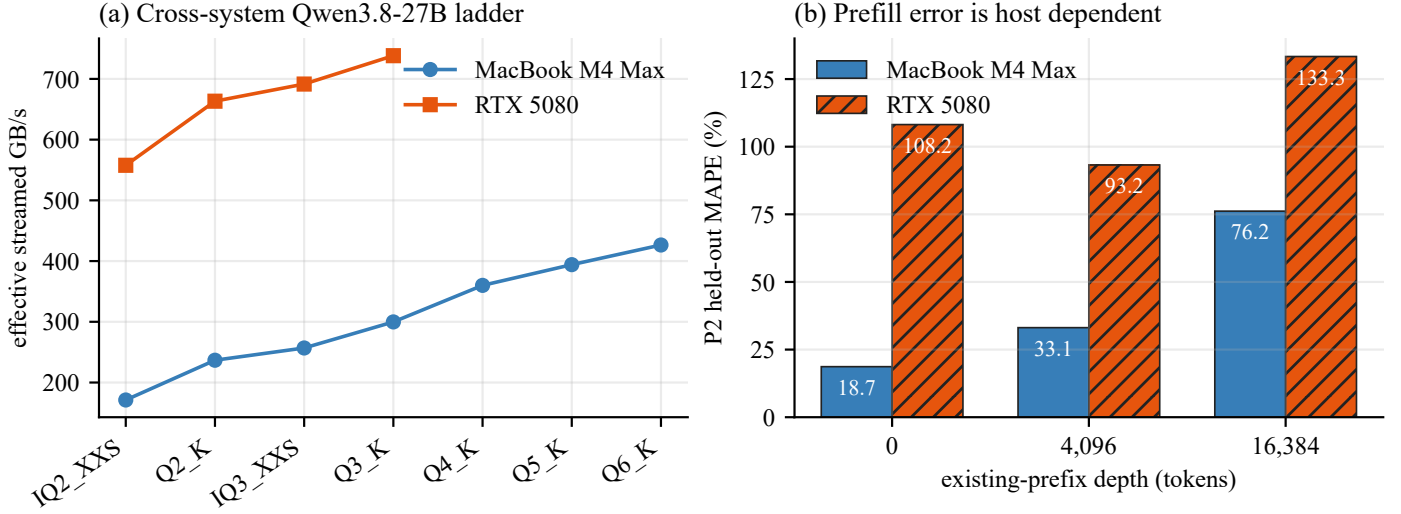


Figure 1: Host-separated diagnostics. Left: effective streamed bandwidth for the scored Qwen3.8-27B quantizations at zero prefix; host coverage differs, and IQ2 has 64 layers/26.90B parameters versus 65 layers/27.32B for the other shown files, so the set is a family comparison rather than a strictly identical-network quantization ladder. Right: P2 held-out error when depth-0 coefficients are applied at each existing-prefix depth. The RTX result is not pooled with the Apple result.

Host	Split	Depth	n	P1 MAPE	P2 MAPE	P2 median	P2 max
MacBook M4 Max	train	0	15	6.81	4.13	0.00	25.99
MacBook M4 Max	train	4,096	15	22.35	20.83	17.58	49.82
MacBook M4 Max	train	16,384	15	66.94	64.13	51.02	150.97
MacBook M4 Max	test	0	4	22.68	18.68	14.90	42.85
MacBook M4 Max	test	4,096	4	40.98	33.11	29.60	55.36
MacBook M4 Max	test	16,384	4	86.59	76.18	73.78	132.42
RTX 5080	train	0	12	14.46	5.86	0.00	25.36
RTX 5080	train	4,096	12	19.26	12.35	8.23	32.74
RTX 5080	train	16,384	12	36.13	28.56	18.19	72.26
RTX 5080	test	0	2	124.51	108.18	108.18	204.07
RTX 5080	test	4,096	2	108.39	93.23	93.23	167.57
RTX 5080	test	16,384	2	151.64	133.34	133.34	227.36

3 Selected model metadata

Tables 2 and 3 jointly reproduce every frozen metadata field used for the 22 unique selected files (21 scored and 1 probe-only). Scored host coverage is explicit because the two measurement cohorts are not identical.

Table 2: Selected model identity, source, and file size. Sizes are exact bytes; host codes are MB (MacBook M4 Max) and RTX (RTX 5080), and “probe only” denotes a file excluded from every prediction fit and score.

ID	Split	Scored host(s)	GGUF filename	Repository	Architecture	Quant.	Bytes
M01	train	MB, RTX	gemma-4-12b-it-qat-q4_0.gguf	google/gemma-4-12b-it-qat-q4_0-gguf	gemma4	Q4_0	6,975,879,296
M02	train	MB	gemma-4-26B-A4B-it-UD-Q4_K_M.gguf	unsloth/gemma-4-26B-A4B-it-GGUF	gemma4	Q4_K_M	16,947,541,728
M03	train	MB, RTX	gpt-oss-20b-MXFP4.gguf	ggml-org/gpt-oss-20b-GGUF	gpt-oss	MXFP4	12,109,566,624
M04	test	MB, RTX	inclusionAI_Ling-mini-2.0-Q4_K_M.gguf	bartowski/inclusionAI_Ling-mini-2.0-GGUF	bailingmoe2	Q4_K_M	9,941,894,336
M05	test	MB	Muse-Glimmer-30B-UD-Q4_K_XL.gguf	unsloth/Muse-Glimmer-30B-GGUF	muse-glimmer	Q4_K	15,878,222,368
M06	test	MB	Nemotron-3-Nano-30B-A3B-Q4_K_M.gguf	unsloth/Nemotron-3-Nano-30B-A3B-GGUF	nemotron_h_moe	Q4_K_M	24,574,373,664
M07	test	MB, RTX	NVIDIA-Nemotron-3-Nano-4B-Q4_K_M.gguf	lmstudio-community/ NVIDIA-Nemotron-3-Nano-4B-GGUF	nemotron_h	Q4_K_M	2,837,072,896
M08	train	MB, RTX	Qwen3.5-4B-Q4_K_M.gguf	unsloth/Qwen3.5-4B-GGUF	qwen35	Q4_K_M	2,740,937,888
M09	train	RTX	Qwen3.5-4B-Q8_0.gguf	unsloth/Qwen3.5-4B-GGUF	qwen35	Q8_0	4,482,403,488
M10	train	MB, RTX	Qwen3.5-9B-Q4_K_M.gguf	unsloth/Qwen3.5-9B-GGUF	qwen35	Q4_K_M	5,680,522,464
M11	train	RTX	Qwen3.5-9B-Q8_0.gguf	unsloth/Qwen3.5-9B-GGUF	qwen35	Q8_0	9,527,502,048
M12	train	MB	Qwen3.6-35B-A3B-UD-Q4_K_M.gguf	unsloth/Qwen3.6-35B-A3B-GGUF	qwen35moe	Q4_K_M	22,134,528,992
M13	probe only	–	Qwen3.8-27B-Q8_0.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q8_0	29,047,086,048
M14	train	MB, RTX	Qwen3.8-27B-UD-IQ2_XXS.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	IQ2_XXS	7,266,070,528
M15	train	MB, RTX	Qwen3.8-27B-UD-IQ3_XXS.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	IQ3_XXS	10,934,860,704
M16	train	MB, RTX	Qwen3.8-27B-UD-Q2_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q2_K	9,828,981,664
M17	train	MB, RTX	Qwen3.8-27B-UD-Q3_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q3_K	13,146,393,504
M18	train	MB	Qwen3.8-27B-UD-Q4_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q4_K	17,559,178,144
M19	train	MB	Qwen3.8-27B-UD-Q5_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q5_K	20,876,938,144
M20	train	MB	Qwen3.8-27B-UD-Q6_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q6_K	25,299,061,664
M21	train	MB, RTX	SmolLM3-Q4_K_M.gguf	ggml-org/SmolLM3-3B-GGUF	smollm3	Q4_K_M	1,915,305,312
M22	train	MB, RTX	SmolLM3-Q8_0.gguf	ggml-org/SmolLM3-3B-GGUF	smollm3	Q8_0	3,275,574,624

Table 3: Selected model geometry. KV columns are exact bytes at the measured depths; k/E is routed/total experts.

ID	Total P	Active P	Active %	Layers	Width	Vocab.	KV heads	Head dim.	Used/all	KV@0	KV@4096	KV@16384
M01	11,907,350,576	11,907,350,576	100.0	48	3,840	262,144	8	512	–	0	738,197,504	939,524,096
M02	25,233,142,046	3,822,527,246	15.1	30	2,816	262,144	8	512	8/128	0	503,316,480	754,974,720
M03	20,914,757,184	4,187,440,704	20.0	24	2,880	201,088	8	64	4/32	0	201,326,592	805,306,368
M04	16,255,643,392	1,432,973,056	8.8	20	2,048	157,184	4	128	8/256	0	167,772,160	671,088,640
M05	27,854,794,240	27,854,794,240	100.0	52	6,656	202,048	2	128	–	0	218,103,808	872,415,232
M06	31,577,940,288	3,580,076,352	11.3	52	2,688	131,072	2	128	6/128	397,934,592	423,100,416	498,597,888
M07	3,973,556,832	3,973,556,832	100.0	42	3,136	131,072	8	128	–	616,366,080	683,474,944	884,801,536
M08	4,205,751,296	4,205,751,296	100.0	32	2,560	248,320	4	256	–	0	536,870,912	2,147,483,648
M09	4,205,751,296	4,205,751,296	100.0	32	2,560	248,320	4	256	–	0	536,870,912	2,147,483,648
M10	8,953,803,264	8,953,803,264	100.0	32	4,096	248,320	4	256	–	0	536,870,912	2,147,483,648
M11	8,953,803,264	8,953,803,264	100.0	32	4,096	248,320	4	256	–	0	536,870,912	2,147,483,648
M12	34,660,610,688	3,454,988,928	10.0	40	2,048	248,320	2	256	8/256	0	335,544,320	1,342,177,280
M13	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M14	26,895,998,464	26,895,998,464	100.0	64	5,120	248,320	4	256	–	0	1,073,741,824	4,294,967,296
M15	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M16	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M17	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M18	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M19	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M20	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M21	3,075,098,624	3,075,098,624	100.0	36	2,048	128,256	4	128	–	0	301,989,888	1,207,959,552
M22	3,075,098,624	3,075,098,624	100.0	36	2,048	128,256	4	128	–	0	301,989,888	1,207,959,552

4 All scored decode observations

These are the 99 rows exported by results/predictions.csv. Throughput is tokens/s; APE is absolute percentage error. Rows are ordered by host, split, model ID, and depth.

Table 4: Protocol-complete decode measurements and B2 predictions. The retry gate uses the worst decode CV in each host–file invocation. Pre-load is the macOS one-minute POSIX load average or, on Windows, busy-logical-CPU equivalents; it is meaningful within a host but not comparable across hosts.

Host	ID	Split	Depth	Measured	B2	APE %	Row CV %	Load	Attempts
MacBook M4 Max	M01	train	0	54.28	54.28	0.00	0.42	15.79	1
MacBook M4 Max	M01	train	4,096	49.91	49.08	1.65	0.55	15.79	1
MacBook M4 Max	M01	train	16,384	45.34	47.84	5.50	1.54	15.79	1
MacBook M4 Max	M02	train	0	89.15	124.01	39.11	0.15	16.58	1
MacBook M4 Max	M02	train	4,096	82.91	103.69	25.07	0.42	16.58	1
MacBook M4 Max	M02	train	16,384	75.37	95.83	27.15	0.44	16.58	1
MacBook M4 Max	M03	train	0	124.28	128.17	3.13	0.29	11.83	1
MacBook M4 Max	M03	train	4,096	118.34	118.34	0.00	0.19	11.83	1
MacBook M4 Max	M03	train	16,384	102.83	96.21	6.43	0.67	11.83	1
MacBook M4 Max	M08	train	0	101.64	116.16	14.29	0.16	8.19	1
MacBook M4 Max	M08	train	4,096	97.13	97.13	0.00	0.28	8.19	1
MacBook M4 Max	M08	train	16,384	85.05	65.13	23.42	1.25	8.19	1
MacBook M4 Max	M10	train	0	66.64	56.05	15.90	1.00	8.30	1
MacBook M4 Max	M10	train	4,096	62.95	51.21	18.65	1.81	8.30	1
MacBook M4 Max	M10	train	16,384	56.62	40.67	28.17	2.81	8.30	1
MacBook M4 Max	M12	train	0	80.85	144.30	78.49	0.29	15.62	1
MacBook M4 Max	M12	train	4,096	78.38	125.26	59.81	0.24	15.62	1
MacBook M4 Max	M12	train	16,384	72.10	89.72	24.44	1.12	15.62	1
MacBook M4 Max	M14	train	0	23.56	23.56	0.00	1.20	16.52	1
MacBook M4 Max	M14	train	4,096	20.41	20.53	0.59	2.70	16.52	1
MacBook M4 Max	M14	train	16,384	19.71	14.81	24.88	1.72	16.52	1
MacBook M4 Max	M15	train	0	23.49	24.42	3.94	1.09	14.84	1
MacBook M4 Max	M15	train	4,096	22.20	22.20	0.00	1.59	14.84	1
MacBook M4 Max	M15	train	16,384	19.63	17.45	11.07	1.65	14.84	1
MacBook M4 Max	M16	train	0	24.08	25.71	6.75	1.90	11.04	1
MacBook M4 Max	M16	train	4,096	23.14	23.14	0.00	2.39	11.04	1
MacBook M4 Max	M16	train	16,384	21.30	17.81	16.41	1.65	11.04	1
MacBook M4 Max	M17	train	0	22.80	23.29	2.12	2.60	11.42	3
MacBook M4 Max	M17	train	4,096	21.89	21.50	1.75	0.91	11.42	3
MacBook M4 Max	M17	train	16,384	17.49	17.49	0.00	2.94	11.42	3
MacBook M4 Max	M18	train	0	20.50	20.92	2.05	2.11	13.06	3

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Table 4: Protocol-complete decode measurements and B2 predictions. The retry gate uses the worst decode CV in each host–file invocation. Pre-load is the macOS one-minute POSIX load average or, on Windows, busy-logical-CPU equivalents; it is meaningful within a host but not comparable across hosts. (continued)

Host	ID	Split	Depth	Measured	B2	APE %	Row CV %	Load	Attempts
MacBook M4 Max	M18	train	4,096	19.70	19.70	0.00	1.97	13.06	3
MacBook M4 Max	M18	train	16,384	16.77	16.76	0.10	1.47	13.06	3
MacBook M4 Max	M19	train	0	18.88	19.13	1.31	1.57	17.66	1
MacBook M4 Max	M19	train	4,096	18.33	18.18	0.83	2.21	17.66	1
MacBook M4 Max	M19	train	16,384	15.82	15.82	0.00	2.67	17.66	1
MacBook M4 Max	M20	train	0	16.86	16.86	0.00	0.37	17.18	1
MacBook M4 Max	M20	train	4,096	15.92	16.16	1.50	1.42	17.18	1
MacBook M4 Max	M20	train	16,384	14.97	14.38	3.96	0.92	17.18	1
MacBook M4 Max	M21	train	0	169.50	166.23	1.93	0.46	13.91	1
MacBook M4 Max	M21	train	4,096	149.47	143.59	3.93	0.31	13.91	1
MacBook M4 Max	M21	train	16,384	115.22	101.94	11.53	0.40	13.91	1
MacBook M4 Max	M22	train	0	118.87	119.53	0.55	0.40	13.92	1
MacBook M4 Max	M22	train	4,096	109.44	109.44	0.00	0.41	13.92	1
MacBook M4 Max	M22	train	16,384	88.22	87.32	1.01	1.00	13.92	1
MacBook M4 Max	M04	test	0	242.33	363.29	49.92	1.24	11.64	1
MacBook M4 Max	M04	test	4,096	219.67	304.92	38.81	0.84	11.64	1
MacBook M4 Max	M04	test	16,384	175.60	205.75	17.17	0.37	11.64	1
MacBook M4 Max	M05	test	0	26.23	23.14	11.81	1.92	8.93	3
MacBook M4 Max	M05	test	4,096	23.30	22.82	2.05	2.13	8.93	3
MacBook M4 Max	M05	test	16,384	22.55	21.93	2.74	3.03	8.93	3
MacBook M4 Max	M06	test	0	92.12	100.00	8.55	0.30	10.01	1
MacBook M4 Max	M06	test	4,096	90.68	99.21	9.41	0.49	10.01	1
MacBook M4 Max	M06	test	16,384	85.16	96.93	13.83	0.68	10.01	1
MacBook M4 Max	M07	test	0	93.29	92.19	1.18	0.30	8.78	1
MacBook M4 Max	M07	test	4,096	91.41	90.44	1.07	0.10	8.78	1
MacBook M4 Max	M07	test	16,384	84.86	85.55	0.80	0.42	8.78	1
RTX 5080	M01	train	0	98.27	104.03	5.86	0.31	3.58	1
RTX 5080	M01	train	4,096	94.08	94.08	0.00	0.64	3.58	1
RTX 5080	M01	train	16,384	92.53	91.68	0.91	0.45	3.58	1
RTX 5080	M03	train	0	229.65	236.51	2.99	0.48	8.58	2
RTX 5080	M03	train	4,096	218.37	218.37	0.00	0.54	8.58	2
RTX 5080	M03	train	16,384	203.03	177.54	12.56	1.58	8.58	2
RTX 5080	M08	train	0	202.01	241.17	19.39	0.33	1.02	1
RTX 5080	M08	train	4,096	198.97	201.67	1.36	0.50	1.02	1
RTX 5080	M08	train	16,384	181.22	135.22	25.38	0.52	1.02	1
RTX 5080	M09	train	0	147.62	163.32	10.64	0.15	1.60	1

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Table 4: Protocol-complete decode measurements and B2 predictions. The retry gate uses the worst decode CV in each host–file invocation. Pre-load is the macOS one-minute POSIX load average or, on Windows, busy-logical-CPU equivalents; it is meaningful within a host but not comparable across hosts. (continued)

Host	ID	Split	Depth	Measured	B2	APE %	Row CV %	Load	Attempts
RTX 5080	M09	train	4,096	145.85	145.85	0.00	0.50	1.60	1
RTX 5080	M09	train	16,384	136.04	110.42	18.83	0.34	1.60	1
RTX 5080	M10	train	0	132.95	116.37	12.47	0.15	1.66	1
RTX 5080	M10	train	4,096	131.63	106.32	19.23	0.43	1.66	1
RTX 5080	M10	train	16,384	123.69	84.44	31.73	0.43	1.66	1
RTX 5080	M11	train	0	88.98	76.84	13.65	0.09	1.47	1
RTX 5080	M11	train	4,096	88.23	72.74	17.55	0.23	1.47	1
RTX 5080	M11	train	16,384	84.85	62.70	26.10	0.24	1.47	1
RTX 5080	M14	train	0	76.76	86.74	13.01	0.33	11.26	2
RTX 5080	M14	train	4,096	75.58	75.58	0.00	0.49	11.26	2
RTX 5080	M14	train	16,384	70.36	54.52	22.51	1.39	11.26	2
RTX 5080	M15	train	0	63.24	68.69	8.62	0.28	14.78	1
RTX 5080	M15	train	4,096	62.46	62.46	0.00	0.41	14.78	1
RTX 5080	M15	train	16,384	58.63	49.10	16.25	1.50	14.78	1
RTX 5080	M16	train	0	67.48	73.66	9.15	0.30	12.10	1
RTX 5080	M16	train	4,096	66.30	66.30	0.00	0.41	12.10	1
RTX 5080	M16	train	16,384	62.22	51.02	18.00	1.65	12.10	1
RTX 5080	M17	train	0	56.15	60.10	7.02	0.22	7.74	1
RTX 5080	M17	train	4,096	55.49	55.49	0.00	0.36	7.74	1
RTX 5080	M17	train	16,384	52.65	45.12	14.30	1.02	7.74	1
RTX 5080	M21	train	0	306.16	345.13	12.73	0.49	1.41	1
RTX 5080	M21	train	4,096	272.17	298.13	9.54	0.74	1.41	1
RTX 5080	M21	train	16,384	211.65	211.65	0.00	0.48	1.41	1
RTX 5080	M22	train	0	209.46	223.49	6.70	0.19	0.70	1
RTX 5080	M22	train	4,096	192.60	204.63	6.24	0.33	0.70	1
RTX 5080	M22	train	16,384	160.88	163.28	1.49	0.29	0.70	1
RTX 5080	M04	test	0	446.42	754.26	68.96	0.76	2.05	1
RTX 5080	M04	test	4,096	406.19	633.07	55.85	1.35	2.05	1
RTX 5080	M04	test	16,384	326.38	427.17	30.88	1.04	2.05	1
RTX 5080	M07	test	0	239.52	191.41	20.09	0.41	1.86	2
RTX 5080	M07	test	4,096	236.79	187.76	20.70	0.29	1.86	2
RTX 5080	M07	test	16,384	223.15	177.61	20.41	1.01	1.86	2

5 All scored prefill observations

These are the 99 rows exported by results/predictions_prefill.csv. P1 has one host coefficient; P2 has one host-by-quantization coefficient. Both are fit only at depth 0. The row CV column is reported rather than used as an inclusion filter.

Table 5: Protocol-complete prefill measurements and predictions. The retry decision was decode-gated, not prefill-gated; depth-0 rows are the primary evaluation and larger depths are scope diagnostics.

Host	ID	Split	Depth	Scope	Measured	P1	P1 APE	P2	P2 APE	Row CV %
MacBook M4 Max	M01	train	0	primary	589.73	547.32	7.19	589.73	0.00	0.05
MacBook M4 Max	M01	train	4,096	diagnostic	491.90	547.32	11.27	589.73	19.89	1.01
MacBook M4 Max	M01	train	16,384	diagnostic	382.76	547.32	42.99	589.73	54.07	3.47
MacBook M4 Max	M02	train	0	primary	1418.29	1704.93	20.21	1610.29	13.54	0.38
MacBook M4 Max	M02	train	4,096	diagnostic	1145.69	1704.93	48.81	1610.29	40.55	2.81
MacBook M4 Max	M02	train	16,384	diagnostic	741.02	1704.93	130.08	1610.29	117.31	2.63
MacBook M4 Max	M03	train	0	primary	1589.84	1556.35	2.11	1589.84	0.00	0.27
MacBook M4 Max	M03	train	4,096	diagnostic	1372.89	1556.35	13.36	1589.84	15.80	2.79
MacBook M4 Max	M03	train	16,384	diagnostic	848.96	1556.35	83.33	1589.84	87.27	0.72
MacBook M4 Max	M08	train	0	primary	1463.56	1549.58	5.88	1463.56	0.00	0.07
MacBook M4 Max	M08	train	4,096	diagnostic	1310.45	1549.58	18.25	1463.56	11.68	0.40
MacBook M4 Max	M08	train	16,384	diagnostic	941.78	1549.58	64.54	1463.56	55.40	0.89
MacBook M4 Max	M10	train	0	primary	819.16	727.86	11.14	687.46	16.08	0.13
MacBook M4 Max	M10	train	4,096	diagnostic	727.07	727.86	0.11	687.46	5.45	0.39
MacBook M4 Max	M10	train	16,384	diagnostic	564.60	727.86	28.92	687.46	21.76	1.59
MacBook M4 Max	M12	train	0	primary	1414.02	1886.30	33.40	1781.59	25.99	0.67
MacBook M4 Max	M12	train	4,096	diagnostic	1189.17	1886.30	58.62	1781.59	49.82	1.38
MacBook M4 Max	M12	train	16,384	diagnostic	880.48	1886.30	114.24	1781.59	102.34	1.39
MacBook M4 Max	M14	train	0	primary	239.52	242.31	1.17	239.52	0.00	2.73
MacBook M4 Max	M14	train	4,096	diagnostic	196.04	242.31	23.60	239.52	22.18	0.31
MacBook M4 Max	M14	train	16,384	diagnostic	158.60	242.31	52.78	239.52	51.02	1.33
MacBook M4 Max	M15	train	0	primary	238.54	238.54	0.00	238.54	0.00	2.42
MacBook M4 Max	M15	train	4,096	diagnostic	203.40	238.54	17.28	238.54	17.28	1.81
MacBook M4 Max	M15	train	16,384	diagnostic	172.91	238.54	37.96	238.54	37.96	1.74
MacBook M4 Max	M16	train	0	primary	235.14	238.54	1.45	235.14	0.00	1.72
MacBook M4 Max	M16	train	4,096	diagnostic	207.61	238.54	14.90	235.14	13.26	2.63
MacBook M4 Max	M16	train	16,384	diagnostic	180.12	238.54	32.43	235.14	30.54	1.42
MacBook M4 Max	M17	train	0	primary	241.51	238.54	1.23	241.51	0.00	2.59
MacBook M4 Max	M17	train	4,096	diagnostic	202.71	238.54	17.67	241.51	19.14	0.93
MacBook M4 Max	M17	train	16,384	diagnostic	164.64	238.54	44.89	241.51	46.69	2.76
MacBook M4 Max	M18	train	0	primary	224.91	238.54	6.06	224.91	0.00	3.91
MacBook M4 Max	M18	train	4,096	diagnostic	191.29	238.54	24.70	224.91	17.58	0.80

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Table 5: Protocol-complete prefill measurements and predictions. The retry decision was decode-gated, not prefill-gated; depth-0 rows are the primary evaluation and larger depths are scope diagnostics. (continued)

Host	ID	Split	Depth	Scope	Measured	P1	P1 APE	P2	P2 APE	Row CV %
MacBook M4 Max	M18	train	16,384	diagnostic	162.37	238.54	46.91	224.91	38.52	2.61
MacBook M4 Max	M19	train	0	primary	225.20	238.54	5.92	225.20	0.00	2.87
MacBook M4 Max	M19	train	4,096	diagnostic	196.00	238.54	21.70	225.20	14.90	2.15
MacBook M4 Max	M19	train	16,384	diagnostic	174.79	238.54	36.47	225.20	28.84	1.41
MacBook M4 Max	M20	train	0	primary	239.03	238.54	0.20	239.03	0.00	2.54
MacBook M4 Max	M20	train	4,096	diagnostic	208.07	238.54	14.64	239.03	14.88	2.22
MacBook M4 Max	M20	train	16,384	diagnostic	181.38	238.54	31.52	239.03	31.78	1.35
MacBook M4 Max	M21	train	0	primary	2135.94	2119.33	0.78	2001.68	6.29	0.04
MacBook M4 Max	M21	train	4,096	diagnostic	1653.76	2119.33	28.15	2001.68	21.04	2.82
MacBook M4 Max	M21	train	16,384	diagnostic	965.07	2119.33	119.60	2001.68	107.41	1.34
MacBook M4 Max	M22	train	0	primary	2239.31	2119.33	5.36	2239.31	0.00	0.12
MacBook M4 Max	M22	train	4,096	diagnostic	1735.34	2119.33	22.13	2239.31	29.04	0.12
MacBook M4 Max	M22	train	16,384	diagnostic	892.27	2119.33	137.52	2239.31	150.97	1.34
MacBook M4 Max	M04	test	0	primary	4208.20	4547.98	8.07	4295.53	2.08	0.42
MacBook M4 Max	M04	test	4,096	diagnostic	3300.39	4547.98	37.80	4295.53	30.15	0.62
MacBook M4 Max	M04	test	16,384	diagnostic	1848.16	4547.98	146.08	4295.53	132.42	1.94
MacBook M4 Max	M05	test	0	primary	236.65	233.97	1.13	220.60	6.78	3.52
MacBook M4 Max	M05	test	4,096	diagnostic	187.15	233.97	25.01	220.60	17.87	3.95
MacBook M4 Max	M05	test	16,384	diagnostic	176.87	233.97	32.28	220.60	24.72	1.62
MacBook M4 Max	M06	test	0	primary	1203.59	1820.39	51.25	1719.34	42.85	0.45
MacBook M4 Max	M06	test	4,096	diagnostic	1106.68	1820.39	64.49	1719.34	55.36	3.64
MacBook M4 Max	M06	test	16,384	diagnostic	895.52	1820.39	103.28	1719.34	91.99	1.19
MacBook M4 Max	M07	test	0	primary	1259.18	1640.13	30.25	1549.08	23.02	4.84
MacBook M4 Max	M07	test	4,096	diagnostic	1200.41	1640.13	36.63	1549.08	29.05	3.30
MacBook M4 Max	M07	test	16,384	diagnostic	995.75	1640.13	64.71	1549.08	55.57	1.37
RTX 5080	M01	train	0	primary	5624.55	4619.99	17.86	5624.55	0.00	5.17
RTX 5080	M01	train	4,096	diagnostic	5030.69	4619.99	8.16	5624.55	11.80	3.92
RTX 5080	M01	train	16,384	diagnostic	4336.80	4619.99	6.53	5624.55	29.69	2.60
RTX 5080	M03	train	0	primary	8633.84	13137.34	52.16	8633.84	0.00	3.75
RTX 5080	M03	train	4,096	diagnostic	8160.96	13137.34	60.98	8633.84	5.79	4.00
RTX 5080	M03	train	16,384	diagnostic	7086.04	13137.34	85.40	8633.84	21.84	2.75
RTX 5080	M08	train	0	primary	9911.27	13080.14	31.97	12128.70	22.37	5.67
RTX 5080	M08	train	4,096	diagnostic	9373.61	13080.14	39.54	12128.70	29.39	4.03
RTX 5080	M08	train	16,384	diagnostic	8143.16	13080.14	60.63	12128.70	48.94	3.62
RTX 5080	M09	train	0	primary	10379.35	13080.14	26.02	13011.54	25.36	5.85
RTX 5080	M09	train	4,096	diagnostic	9802.11	13080.14	33.44	13011.54	32.74	4.92
RTX 5080	M09	train	16,384	diagnostic	8448.91	13080.14	54.81	13011.54	54.00	3.69

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Table 5: Protocol-complete prefill measurements and predictions. The retry decision was decode-gated, not prefill-gated; depth-0 rows are the primary evaluation and larger depths are scope diagnostics. (continued)

Host	ID	Split	Depth	Scope	Measured	P1	P1 APE	P2	P2 APE	Row CV %
RTX 5080	M10	train	0	primary	6452.64	6143.96	4.78	5697.05	11.71	3.37
RTX 5080	M10	train	4,096	diagnostic	6219.18	6143.96	1.21	5697.05	8.40	2.66
RTX 5080	M10	train	16,384	diagnostic	5738.49	6143.96	7.07	5697.05	0.72	2.17
RTX 5080	M11	train	0	primary	6861.31	6143.96	10.45	6111.74	10.92	4.42
RTX 5080	M11	train	4,096	diagnostic	6647.95	6143.96	7.58	6111.74	8.07	2.90
RTX 5080	M11	train	16,384	diagnostic	6079.76	6143.96	1.06	6111.74	0.53	2.86
RTX 5080	M14	train	0	primary	1725.38	2045.35	18.54	1725.38	0.00	3.23
RTX 5080	M14	train	4,096	diagnostic	1682.57	2045.35	21.56	1725.38	2.54	1.97
RTX 5080	M14	train	16,384	diagnostic	1538.25	2045.35	32.97	1725.38	12.17	2.17
RTX 5080	M15	train	0	primary	2024.12	2013.56	0.52	2024.12	0.00	3.50
RTX 5080	M15	train	4,096	diagnostic	1954.57	2013.56	3.02	2024.12	3.56	2.46
RTX 5080	M15	train	16,384	diagnostic	1771.53	2013.56	13.66	2024.12	14.26	2.22
RTX 5080	M16	train	0	primary	2043.07	2013.56	1.44	2043.07	0.00	3.11
RTX 5080	M16	train	4,096	diagnostic	1962.09	2013.56	2.62	2043.07	4.13	3.14
RTX 5080	M16	train	16,384	diagnostic	1783.83	2013.56	12.88	2043.07	14.53	2.23
RTX 5080	M17	train	0	primary	2042.02	2013.56	1.39	2042.02	0.00	17.04
RTX 5080	M17	train	4,096	diagnostic	2114.20	2013.56	4.76	2042.02	3.41	2.90
RTX 5080	M17	train	16,384	diagnostic	1907.04	2013.56	5.59	2042.02	7.08	2.39
RTX 5080	M21	train	0	primary	16588.19	17889.45	7.84	16588.19	0.00	7.30
RTX 5080	M21	train	4,096	diagnostic	14035.04	17889.45	27.46	16588.19	18.19	5.06
RTX 5080	M21	train	16,384	diagnostic	9948.68	17889.45	79.82	16588.19	66.74	3.46
RTX 5080	M22	train	0	primary	17795.63	17889.45	0.53	17795.63	0.00	8.37
RTX 5080	M22	train	4,096	diagnostic	14813.44	17889.45	20.76	17795.63	20.13	5.96
RTX 5080	M22	train	16,384	diagnostic	10330.65	17889.45	73.17	17795.63	72.26	3.83
RTX 5080	M04	test	0	primary	11706.91	38390.00	227.93	35597.53	204.07	39.19
RTX 5080	M04	test	4,096	diagnostic	13303.93	38390.00	188.56	35597.53	167.57	5.45
RTX 5080	M04	test	16,384	diagnostic	10873.99	38390.00	253.04	35597.53	227.36	4.90
RTX 5080	M07	test	0	primary	11433.58	13844.48	21.09	12837.44	12.28	4.86
RTX 5080	M07	test	4,096	diagnostic	10797.04	13844.48	28.22	12837.44	18.90	3.44
RTX 5080	M07	test	16,384	diagnostic	9215.27	13844.48	50.23	12837.44	39.31	3.36

6 Repeatability and calibration

A separate repeatability series exists only for the MacBook M4 Max; RTX 5080 variability is represented by the five within-invocation repetitions in the row tables, not by an equivalent across-run series. The six Mac runs give mean decode throughput 166.457 tokens/s and sample between-run CV 1.613%; prefill is 2133.091 tokens/s with CV 0.373%. The max-minus-min spans are 4.34% and 0.97% of their respective means. Decode has a descriptive linear trend of 0.74% per run. These are single-cell diagnostics, not general detection thresholds.

Run	Timestamp	Settle (s)	Prefill	Decode	Decode within-run SD
1	2026-08-30T01:06:08.096930-07:00	45	2137.572	164.323	6.520
2	2026-08-30T01:06:59.837356-07:00	45	2136.650	165.447	1.473
3	2026-08-30T01:07:51.671530-07:00	45	2136.742	163.200	10.629
4	2026-08-30T01:08:43.573411-07:00	45	2116.939	166.925	2.338
5	2026-08-30T01:09:35.318847-07:00	45	2135.238	168.415	0.805
6	2026-08-30T01:10:26.991582-07:00	45	2135.403	170.432	0.464

The append-only Mac measurement file also preserves a complete pre-protocol decode grid. It is excluded from every fit and score above. The comparison is Mac-only and is not pooled with RTX data.

Pass	All <i>n</i>	Test <i>n</i>	Test MAPE	Median	P90	Max	All-row CV med.	All-row CV max
Archived pre-protocol	57	12	19.30	15.77	25.94	74.59	1.74	14.91
Final Mac protocol-complete	57	12	13.11	8.98	36.65	49.92	1.00	3.03

For comparison, the retained historical uncontrolled file contains two depth-0 observations of the same Qwen3.5-4B Q4_K_M cell:

Phase	Higher throughput	Lower throughput	Decrease (%)
prefill	1467.424	1090.676	25.67
decode	103.616	87.594	15.46

6.1 Host settings and calibration

Host	Platform	Accelerator	Threads	Reps	Settle (s)	Requested ngl
MacBook M4 Max	Darwin-arm64	arm (Metal)	12	5	45	99
RTX 5080	Windows-AMD64	NVIDIA GeForce RTX 5080	16	5	45	99

The requested `n_gpu_layers=99` setting asks the runtime to offload as many layers as it can; it is not direct telemetry proving that every tensor and KV allocation remained resident on the accelerator. No device-memory trace was recorded, so the appendix does not label these observations as confirmed fully resident.

Host	Calibration	Value	Detail
MacBook M4 Max	CPU STREAM-style triad	48.558 GB/s	7 repetitions
MacBook M4 Max	CPU FP32 matrix multiply	2.458 TFLOP/s	5 repetitions
MacBook M4 Max	MPS device copy	380.052 GB/s	arm (Metal)
MacBook M4 Max	MPS FP16 matrix multiply	13.950 TFLOP/s	arm (Metal)
MacBook M4 Max	LLM reference ceiling	377.244 GB/s	Qwen3.5-9B-Q8_0.gguf
RTX 5080	CPU STREAM-style triad	13.458 GB/s	7 repetitions
RTX 5080	CPU FP32 matrix multiply	2.314 TFLOP/s	5 repetitions
RTX 5080	CUDA device copy	801.061 GB/s	NVIDIA GeForce RTX 5080
RTX 5080	CUDA FP16 matrix multiply	118.829 TFLOP/s	NVIDIA GeForce RTX 5080

For the fitted B1/B2 and P1/P2 models, the measured bandwidth or FLOP/s scale cancels algebraically after fitting on this same host. It sets the scale of B0 and supports roofline interpretation; it does not by itself produce the reported fitted-model accuracy. The exported row-level bandwidth field uses the device-copy result on both hosts. The Mac LLM reference is retained as a diagnostic and is not the B0 bandwidth value used in the current exports.

Host	Probe	File GB	Decode	SD	CV (%)	Effective GB/s
MacBook M4 Max	Qwen3.8-27B-Q8_0.gguf	29.05	7.100	4.199	59.14	196.653
MacBook M4 Max	Qwen3.5-9B-Q8_0.gguf	9.53	44.661	0.586	1.31	377.244
MacBook M4 Max	Qwen3.5-4B-Q8_0.gguf	4.48	70.817	5.524	7.80	317.432

7 Output-projection sensitivity

The additive output-projection term is reported as a sensitivity analysis. It is not selected: held-out improvement is not consistent across loss functions and hosts, and the term partly recharges weights already included in the streamed-byte term. Host rows and the unequally weighted all-host aggregate are shown separately. The one-term rows in this section are refit under the listed time-domain losses; they are not the median-ratio B2 headline and need not reproduce its MAPE.

Host	Loss	Model	Split	n	MAPE	Median	P90	Max
MacBook M4 Max	absolute time	one term	train	45	12.38	4.97	34.07	91.71
MacBook M4 Max	absolute time	one term	test	12	19.26	14.43	46.77	61.03
MacBook M4 Max	absolute time	two terms	train	45	9.88	4.47	29.64	62.39
MacBook M4 Max	absolute time	two terms	test	12	34.03	30.21	62.96	72.96
RTX 5080	absolute time	one term	train	36	14.18	11.77	28.14	45.36
RTX 5080	absolute time	one term	test	6	44.01	31.40	97.73	105.71
RTX 5080	absolute time	two terms	train	36	12.84	9.76	30.05	39.86
RTX 5080	absolute time	two terms	test	6	27.30	24.89	55.35	56.39
all hosts	absolute time	one term	train	81	13.18	8.06	33.36	91.71
all hosts	absolute time	one term	test	18	27.51	14.43	69.65	105.71
all hosts	absolute time	two terms	train	81	11.20	5.90	30.18	62.39
all hosts	absolute time	two terms	test	18	31.79	30.21	61.58	72.96
MacBook M4 Max	relative time	one term	train	45	11.40	4.35	28.13	83.22
MacBook M4 Max	relative time	one term	test	12	15.17	11.87	40.27	53.89
MacBook M4 Max	relative time	two terms	train	45	8.34	4.68	18.01	46.86
MacBook M4 Max	relative time	two terms	test	12	20.58	19.27	38.22	45.61
RTX 5080	relative time	one term	train	36	12.55	11.72	22.83	34.79
RTX 5080	relative time	one term	test	6	40.81	29.12	83.35	90.75
RTX 5080	relative time	two terms	train	36	11.80	10.35	20.95	28.62
RTX 5080	relative time	two terms	test	6	28.48	23.66	54.83	57.34
all hosts	relative time	one term	train	81	11.91	8.81	26.26	83.22
all hosts	relative time	one term	test	18	23.72	11.87	60.51	90.75
all hosts	relative time	two terms	train	81	9.87	6.65	20.34	46.86
all hosts	relative time	two terms	test	18	23.22	19.27	47.62	57.34

8 Provenance and explicit limits

The on-disk inputs used for this appendix have the following SHA-256 digests. The repository HEAD at generation was d1b707c2ed4d0b483b0edc6221f1894676e8149e; the generator does not assert a clean working tree, so the file digests, rather than this commit alone, identify the inputs.

Table 6: Exact inputs used to generate this appendix.

Role	Path	SHA-256
measurements (lun-mac)	results/measurements_lun-mac.csv	99dcbf9abb9454877713198909703c1f77e00555f0258f7bf9d83e4395be99b4
measurements (rtx5080)	results/measurements_rtx5080.csv	5f4d46f4952b2dad923628c5256edb8adff77301aac835d252dd07b17393e91
decode predictions	results/predictions.csv	5c0020859871a943a960f1bcf7c96e40bc7abd2005dd817c2fcb9962af6f4066
prefill predictions	results/predictions_prefill.csv	430ca9a302b263c8bec7aae51bfb9d2bbf79182c1c4d9c9c1efbd6712ee9e436
decode aggregate error table	results/error_table.csv	37f654f71d160a3c4676fe685f1a36335998fdd89d317f29e314fcc0975fa4af
decode error table by host	results/error_table_by_host.csv	4dfb6874a0f3fd33bb4d2c747ed4da8d8ff916314003658885304299109f158a
prefill aggregate error table	results/error_table_prefill.csv	067f16c7aa5677a234cf59dfe6fccfc967d62e8348b63014adcfcf952485c2b5
prefill error table by host	results/error_table_prefill_by_host.csv	a5a1b9ba99396f7571b01617b381fdbef1fb8d063c07a42913d633230d5e56692
calibration (lun-mac)	results/calibration_lun-mac.json	1848525fd55119b745d16c51a53f0fa7793c81c6d5c4738af38bb838795ee5a8
calibration (rtx5080)	results/calibration_rtx5080.json	4a23505686da0f75a594fef75ac8047c5f06ff63b2bbd43856c0e18cde631591
environment (lun-mac)	results/env_lun-mac.json	1cda40d02b90d0ca21ab7c9cf02cbf6c36bf920a74168c88a78a1456ab7ef59
environment (rtx5080)	results/env_rtx5080.json	0f695cc20ca99843c58f656e05303377f11d4083f357beadb7da8d72a02c7f9f
manifest	results/model_manifest.json	c24b4d8bfb4c1b27a4204775269e8e02f46936914e4f72117f1c955f600ff5c7
metadata	results/model_metadata.json	cb1afadb7fad3b6d9d2df88b7176d72f4c25068bbb865e8c9a33cc79a3e4a2e5
repeatability (lun-mac)	results/repeatability_lun-mac.csv	f79f9565aebc662bde98d1fefa44b29d73e83188c60b6bf60a1061cc2de9f622
archived uncontrolled comparison	results/contaminated/measurements_UNCONTROLLED_lun-mac.csv.bak	b344118d3902a9aa6530a6113a3c5d0c0310ae2a703ad4986c8146d70821eb50
analysis implementation	llmperf/analyze.py	f94a0b83f4d8cd5a3c7e1bf8da3a8b4121ecd3578191484b8b7e7c39bb6a5374
refinement implementation	llmperf/refine.py	523dd6972e20543c314a4f40761ce5fd75da5d2f6b86532c739d022fed535786
metadata implementation	llmperf/common.py	f1f4a7cd9a927b05344cec77ebad6492b8f8195509f897df5443ca50604df592
appendix generator	paper/icassp2027/generate_supplement.py	65bf1ce03fabf1b7d5b6cb356c3b317ad7e7554cb8c4659c9daef169fb119590

- The MacBook Pro M4 Max and RTX 5080 contributed unequal, overlapping cohorts. The planned Mac Studio has no rows in this dataset. Headline coefficients are fit separately by measured host. The leave-one-host-out audit therefore uses only two systems and does not establish universal transfer to new runtime stacks or formats.
- Every scored row records a requested `n_gpu_layers=99`, but no device-memory trace or runtime-reported resident-layer count was retained. This is a maximal-offload request, not proof of full accelerator residency; no partial-offload sweep or offload-cliff result exists.
- The largest RTX modeled file-plus-KV footprint is 17.508 GB for `Qwen3.8-27B-UD-Q3_K_XL.gguf` at depth 16,384, while calibration reports 17.094 GB of device memory. The differing accounting conventions further preclude a residency claim.
- The Mac rows use the placeholder commit value `no git`; both runtime version queries returned usage lines rather than immutable `llama.cpp` revisions. The RTX project commit field is not itself a runtime binary revision.
- Manifest entries identify repository, filename, and byte size, but not immutable repository revisions or full-file hashes. Exact experimental replay is therefore not guaranteed.
- Derived parameter counts and per-depth KV bytes are consumed from the frozen metadata snapshot during appendix generation. This reproduces the analysis but is not a fresh, independent metadata extraction.
- Held-out coverage is small: 4 Mac and 2 RTX configurations, all Q4-family. Other quantization formats occur only in training data. No learning curve establishes how many reference configurations suffice.
- Decode MAPE treats three depths from each host–file configuration as observations; those rows are correlated. No narrow confidence claim is warranted.
- Prefill’s primary result is restricted to depth 0. Larger-depth predictions are scope diagnostics because the equation has no existing-prefix term. The retry rule is decode-only, and 29 of 42 scored RTX prefill rows exceed 3% within-run CV; the RTX prefill result should therefore be treated as both inaccurate and noisy.
- The three Q8 calibration probes are excluded only from the Mac fit and score; two corresponding files are scored on RTX. The 63.39-GB `gpt-oss-120B` Mac attempt is recorded as a prompt-batch decode failure and contributes no throughput row; it was not attempted on RTX.
- `Qwen3.8-27B IQ2` has 64 layers and 26.90B parameters, whereas the other shown `Qwen3.8-27B` files have 65 layers and 27.32B parameters. The quantization plot is therefore a near-family comparison, not a controlled bit-format substitution for one identical network.

The metadata snapshot additionally records these parser/source Git object IDs:

Object	Identifier
Code commit	1bdce5e0d917dce152cba4809d05fc1922a98797
Measurement blob	0ec72c61f2eaa371ba4f591a28964ec602d0686b
Predictions blob	de86a4286026e3b895b3c3809f9b4b0eb7dbda47
Shape-log blob	fa3d7b9cbf4a731219c7d52829fde031c67e3a2d